UC Name	*UC1: User Authentication and Authorization*
Summary	*Authenticate users and give them access according to their permissions and roles.*
Dependency	*None*
Actors	*Primary Actor: All Users*
Preconditions	*The user tries to access the system.*
Description of the Main Sequence	● *User enters username and password.* ● *System verifies information.* ● *If the information is valid, the system checks the permissions and role of the user.* ● *System grants access to the user.*
Description of the Alternative Sequence	*None*
Non functional requirements	*The authentication process should occur within an acceptable time frame, it should check and validate the information in the most secure way possible. The system should handle simultaneous requests.*
Postconditions	*If the credentials are correct the user will be granted access to the system.*

UC Name	*UC2: User Management*
Summary	*Administrators manage user accounts and profiles*
Dependency	*None*
Actors	*Primary Actor: Administrator* *Secondary Actors: Users*
Preconditions	*Administrator is logged into the system.*
Description of the Main Sequence	● *Administrator creates a new user account, specifying the role and permissions.* ● *Administrator deletes user accounts if necessary.* ● *Administrator provides initial login information, which can be updated by the users.* ● *The users can reset their password without permission from the administrator, but they have to request the creation of an account.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Privacy of user data and security measures*
Postconditions	*User accounts are created or deleted accordingly.*

UC Name	*UC3: Airport Personnel Working Hours and Earnings*
Summary	*The system has a database that stores the number of hours worked by each member of the airport personnel and their earnings*
Dependency	*None*
Actors	*Primary Actor: Airport Personnel Member*
Preconditions	*The airport personnel member is logged into the system.*
Description of the Main Sequence	● *The airport personnel member scans a card when he/she starts his/her shift.* ● *The specific ID is stored in the database, serving as a Primary Key in the Airport_Personnel table.* ● *The total hours worked and the total earnings can be displayed from the database.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Proper time tracking and correct earnings calculation.*
Postconditions	*Airport personnel working hours and earnings are recorded and can be viewed.*

UC Name	*UC4: Security Reporting*
Summary	*Security personnel reports incidents and has access to emergency response systems.*
Dependency	*None*
Actors	*Primary Actor: Security Personnel Member*
Preconditions	*Security personnel are logged into the system*
Description of the Main Sequence	● *Security personnel report incidents with location and description* ● *Security personnel has access to emergency response systems*
Description of the Alternative Sequence	*None*
Non functional requirements	*Quick access to emergency systems and efficient incident reporting.*
Postconditions	*Incidents are recorded and emergency systems are ready for quick response.*

UC Name	*UC5: Management Tools*
Summary	*Managers access certain functionalities and monitor key performance indicators*
Dependency	*None*
Actors	*Primary Actor: Manager*
Preconditions	*Manager is logged into the system*
Description of the Main Sequence	● *Managers access functionalities according to their area of responsibility* ● *Managers view reports for monitoring key performance indicators.*
Description of the Alternative Sequence	*None*
Non functional requirements	*User-friendly interface for performing actions and good data visualization for monitoring key performance indicators*
Postconditions	*Managers access data for decision making.*

UC Name	*UC6: Flight Management*
Summary	*Flight managers monitor different aspects of flight operations*
Dependency	*None*
Actors	*Primary Actor: Flight Manager*
Preconditions	*Flight manager is logged into the system*
Description of the Main Sequence	● *Flight manager schedules and coordinates aircraft movements.* ● *Flight manager assigns gates.* ● *Flight manager coordinates ground operations.* ● *Flight managers ensure the completion of safety protocols and regulations.* ● *Flight manager develops emergency response plans.* ● *Flight manager tracks flight performance metrics.* ● *Flight manager generates reports on flight-related data.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Real-time updates for flight scheduling.*
Postconditions	*Flight operations are managed effectively, according to regulations and flight-related data is recorded and managed properly.*

UC Name	*UC7: Cargo Management*
Summary	*Cargo managers monitor the handling and tracking of cargo shipments.*
Dependency	*None*
Actors	*Primary Actor: Cargo Manager*
Preconditions	*Cargo manager is logged into the system.*
Description of the Main Sequence	● *Cargo manager approves or rejects shipment requests.* ● *Cargo manager views and analyzes cargo statistics.* ● *Cargo manager marks cargo arrival and tracks its location.* ● *Cargo manager can see if it is possible to accept the cargo at a given time and the system will provide information about the factors that affect this process.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Secure handling of sensitive cargo information, real-time tracking updates*
Postconditions	*Cargo shipments are managed effectively, and their status is accurately tracked*

UC Name	*UC8: Financial Management*
Summary	*Finance Managers monitor financial transactions and analyze financial data.*
Dependency	*None*
Actors	*Primary Actor: Finance Manager*
Preconditions	*Finance manager is logged into the system.*
Description of the Main Sequence	● *Finance manager accesses transaction data.* ● *Finance manager views and analyzes transaction statistics.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Data encryption for financial transactions, robust data analysis tools*
Postconditions	*Financial transactions are recorded accurately, and financial data is available for analysis*

UC Name	*UC9: Administrator Role*
Summary	*Administrators monitor user management and overall system performance*
Dependency	*None*
Actors	*Primary Actor: Administrator*
Preconditions	*Administrator is logged into the system.*
Description of the Main Sequence	● *Administrator creates new user accounts with assigned roles.* ● *Administrator monitors overall airport performance across various departments.* ● *Administrator accesses information on flights, passengers, security, and staff finance.* ● *Administrator performs all functionalities available to managers.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Admin privileges securely managed, system performance monitoring tools*
Postconditions	*User accounts are managed effectively, and system performance is monitored*

UC Name	*UC10: Owner Role*
Summary	*Owners have full system access and can view all relevant information*
Dependency	*None*
Actors	*Primary Actor: Owner*
Preconditions	*Owner is logged into the system.*
Description of the Main Sequence	● *Owner views total costs, revenue, and profits.* ● *Owner accesses all information available to administrators.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Good data visualization and high-level data access tools*
Postconditions	*Owner has access to all system information*

UC Name	*UC11: Client User Interface*
Summary	*Clients view general flight information and perform certain actions without logging in.*
Dependency	*None*
Actors	*Primary Actor: Client*
Preconditions	*Client accesses the homepage of the system's website (a section made specifically for clients)*
Description of the Main Sequence	● *Client views general flight information.* ● *The system allows clients to perform other action such as: passenger processing, including self-service check-in kiosks, biometric authentication for border control, baggage handling and tracking, and self-boarding*
Description of the Alternative Sequence	*None*
Non functional requirements	*User-friendly interface that is easy to navigate through*
Postconditions	*Clients obtain flight information without having an account and perform certain actions*

UC Name	*UC12: User Interface Customization*
Summary	*The system allows users to customize and choose their interface design preferences*
Dependency	*None*
Actors	*Primary Actor: All Users*
Preconditions	*User is logged into the system*
Description of the Main Sequence	● *User accesses interface customization settings.* ● *User chooses interface design preferences (theme, layout).*
Description of the Alternative Sequence	*None*
Non functional requirements	*Customization helps improve user experience*
Postconditions	*The interface reflects the customization preferences selected by the user.*

UC Name	*UC13: Integration with External Systems*
Summary	*Enable data exchange with outer systems.*
Dependency	*None*
Actors	*Primary Actor: Input/Output Device Actor*
Preconditions	*System works and is operational*
Description of the Main Sequence	● *System establishes connections and exchanges data with external systems.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Secure and reliable data exchange mechanisms*
Postconditions	*System successfully integrates with external systems*

UC Name	*UC14: System Backup and Recovery*
Summary	*Implement backup and recovery mechanisms for system data*
Dependency	*None*
Actors	*Primary Actor: Timer Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System regularly backs up data to secure storage.* ● *System tests backup integrity and performs recovery drills.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Reliable backup and recovery procedures*
Postconditions	*System data is backed up securely and recoverable in case of data loss*

UC Name	*UC15: Performance Optimization*
Summary	*Optimize system performance for efficiency and responsiveness*
Dependency	*None*
Actors	*Primary Actor: Input/Output Device Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System analyzes performance metrics to identify bottlenecks.* ● *System implements optimizations to improve speed and responsiveness.*
Description of the Alternative Sequence	*None*
Non functional requirements	*System response time should meet defined performance benchmarks*
Postconditions	*System performance is optimized for efficient operation*

UC Name	*UC16: Regulatory Compliance*
Summary	*Ensure system compliance with relevant regulations and standards*
Dependency	*UC5: Management Tools*
Actors	*Primary Actor: Input/Output Device Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System identifies applicable regulations and standards.* ● *System implements necessary measures to ensure compliance.*
Description of the Alternative Sequence	*None*
Non functional requirements	*System updates and adapts to changes in regulatory requirements*
Postconditions	*System operates in accordance with regulatory standards*

UC Name	*UC17: Continuous System Monitoring*
Summary	*Implement continuous monitoring of system health and performance*
Dependency	*UC5: Management Tools*
Actors	*Primary Actor: Timer Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System continuously monitors key performance indicators.* ● *System alerts administrators of any abnormalities or issues.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Real-time monitoring of system status and performance*
Postconditions	*System health and performance are continuously monitored*

UC Name	*UC18: Data Privacy and Security*
Summary	*Ensure data privacy and security measures are implemented*
Dependency	*None*
Actors	*Primary Actor: Input/Output Device Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System encrypts sensitive data to protect privacy.* ● *System implements access controls to restrict data access.* ● *The cybersecurity system ensures that the data is safe and protected.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Compliance with data protection regulations and standards*
Postconditions	*Data privacy and security measures are effectively enforced*

UC Name	*UC19: System Availability*
Summary	*Ensure high availability and uptime of the system*
Dependency	*None*
Actors	*Primary Actor: Timer Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System employs redundancy and failover mechanisms.* ● *System conducts regular maintenance and updates.*
Description of the Alternative Sequence	*None*
Non functional requirements	*System should be available at all times*
Postconditions	*System remains available and accessible to users*

UC Name	*UC20: Invoicing and Billing*
Summary	*The system generates invoices and bills for airport operations*
Dependency	*U8: Financial Management*
Actors	*Primary Actor: Input/Output Device Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System generates bills for aircraft operations, parking, landing and departure fees, ground service fees*
Description of the Alternative Sequence	*None*
Non functional requirements	*Ensure secure, reliable, and efficient billing and invoicing operations within the airport management system*
Postconditions	*The invoicing and billing process is handled properly by the system*

UC Name	*UC21: Task Management*
Summary	*Allow managers to assign tasks and track their completion*
Dependency	*UC5: Management Tools*
Actors	*Primary Actor: Manager*
Preconditions	*Manger is logged into the system.*
Description of the Main Sequence	● *Manager assigns tasks to employees.* ● *System tracks task progress and completion status.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Task prioritization and deadline management*
Postconditions	*Tasks are assigned and tracked efficiently*

UC Name	*UC22: Data Visualization*
Summary	*Provide visual representations of data for better understanding*
Dependency	*UC5: Management Tools* *UC10: Owner Role*
Actors	*Primary Actor: All Users*
Preconditions	*User is logged into the system.*
Description of the Main Sequence	● *System generates visualizations based on data.* ● *Users access and interact with visualizations for insights.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Interactive and informative data visualizations*
Postconditions	*Users gain insights from visual representations of data*

UC Name	*UC23: System Notifications*
Summary	*Send notifications to users for important events and updates*
Dependency	*None*
Actors	*Primary Actor: Input/Output Device Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System generates notifications for relevant events.* ● *System sends notifications to users via preferred channels.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Customizable notification settings for users*
Postconditions	*Users receive timely notifications for important updates*

UC Name	*UC24: System Logs Management*
Summary	*Manage and analyze system logs for troubleshooting and auditing*
Dependency	*UC9: Administrator Role*
Actors	*Primary Actor: Administrator*
Preconditions	*Administrator is logged into the system.*
Description of the Main Sequence	● *Administrator accesses system logs.* ● *Administrator analyzes logs for troubleshooting and auditing purposes.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Efficient log storage and retrieval mechanisms*
Postconditions	*System logs are managed effectively for analysis and auditing*

UC Name	*UC25: System Configuration*
Summary	*Allow administrators to configure system settings and parameters*
Dependency	*UC9: Administrator Tools*
Actors	*Primary Actor: Administrator*
Preconditions	*Administrator is logged into the system.*
Description of the Main Sequence	● *Administrator accesses system configuration settings.* ● *Administrator modifies settings and parameters as needed.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Secure configuration management processes*
Postconditions	*System settings are configured according to administrator requirements*

UC Name	*UC26: Disaster Recovery Planning*
Summary	*Develop and implement plans for system recovery in case of disasters*
Dependency	*UC14: System Backup and Recovery*
Actors	*Primary Actor: Timer Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System identifies potential disaster scenarios.* ● *System develops recovery plans for each scenario.* ● *System conducts drills and tests for recovery procedures.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Robust disaster recovery plans and procedures*
Postconditions	*System is prepared to recover from disasters effectively*

UC Name	*UC27: Flight Demand Forecasting*
Summary	*Implement predictive analytics to forecast flight demand*
Dependency	*UC6: Flight Management*
Actors	*Primary Actor: Flight Manager*
Preconditions	*Historical flight data and passenger statistics available.*
Description of the Main Sequence	● *Flight manager selects relevant datasets for analysis, including historical flight schedules and passenger demographics.* ● *System applies predictive models to analyze historical data and predict future flight demand patterns.* ● *System generates forecasts for flight demand based on analyzed data, considering factors such as seasonality, holidays, and special events.*
Description of the Alternative Sequence	*None*
Non functional requirements	*Accurate prediction models tailored to airport operations*
Postconditions	*Flight demand forecasts enable efficient resource allocation and scheduling*

UC Name	*UC28: User Feedback and Improvement*
Summary	*Gather user feedback and suggestions for system improvement*
Dependency	*None*
Actors	*Primary Actor: Input/Output Device Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System provides channels for users to submit feedback and suggestions.* ● *System collects and analyzes user feedback.* ● *System implements improvements based on user input.*
Description of the Alternative Sequence	*None*
Non functional requirements	*User-friendly feedback submission mechanisms*
Postconditions	*System incorporates user feedback for continuous improvement*

UC Name	*UC29: Real-Time Flight Updates*
Summary	*Provide real-time updates for flight statuses and information*
Dependency	*UC6: Flight Management*
Actors	*Primary Actor: Input/Output Device Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System continuously monitors flight status data.* ● *Upon changes, system updates flight information in real-time.*
Description of the Alternative Sequence	• *If there's a delay in receiving updates, system notifies users of potential data lag.*
Non functional requirements	*Reliable data streaming for real-time updates*
Postconditions	*Users have access to up-to-date flight information in real-time*

UC Name	*UC30: Emergency Response Coordination*
Summary	*Coordinate emergency response activities efficiently*
Dependency	*UC4: Security Management*
Actors	*Primary Actor: Input/Output Device Actor*
Preconditions	*System works and is operational.*
Description of the Main Sequence	● *System receives notification of an emergency incident.* ● *System identifies appropriate response protocols based on incident type.* ● *System notifies relevant personnel and authorities about the incident.* ● *System tracks and coordinates response efforts in real-time.*
Description of the Alternative Sequence	*If the incident type is unclear, system prompts for additional information or escalates to a higher authority.*
Non functional requirements	*Seamless communication channels for rapid response*
Postconditions	*Emergency response activities are coordinated effectively*